Derivative as a Function
(Section 3.2)

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MATH 2413

The Definitions of the Derivative

Definition

The **derivative** of a function $f(x)$ at the point is also a function and can be calculated as:

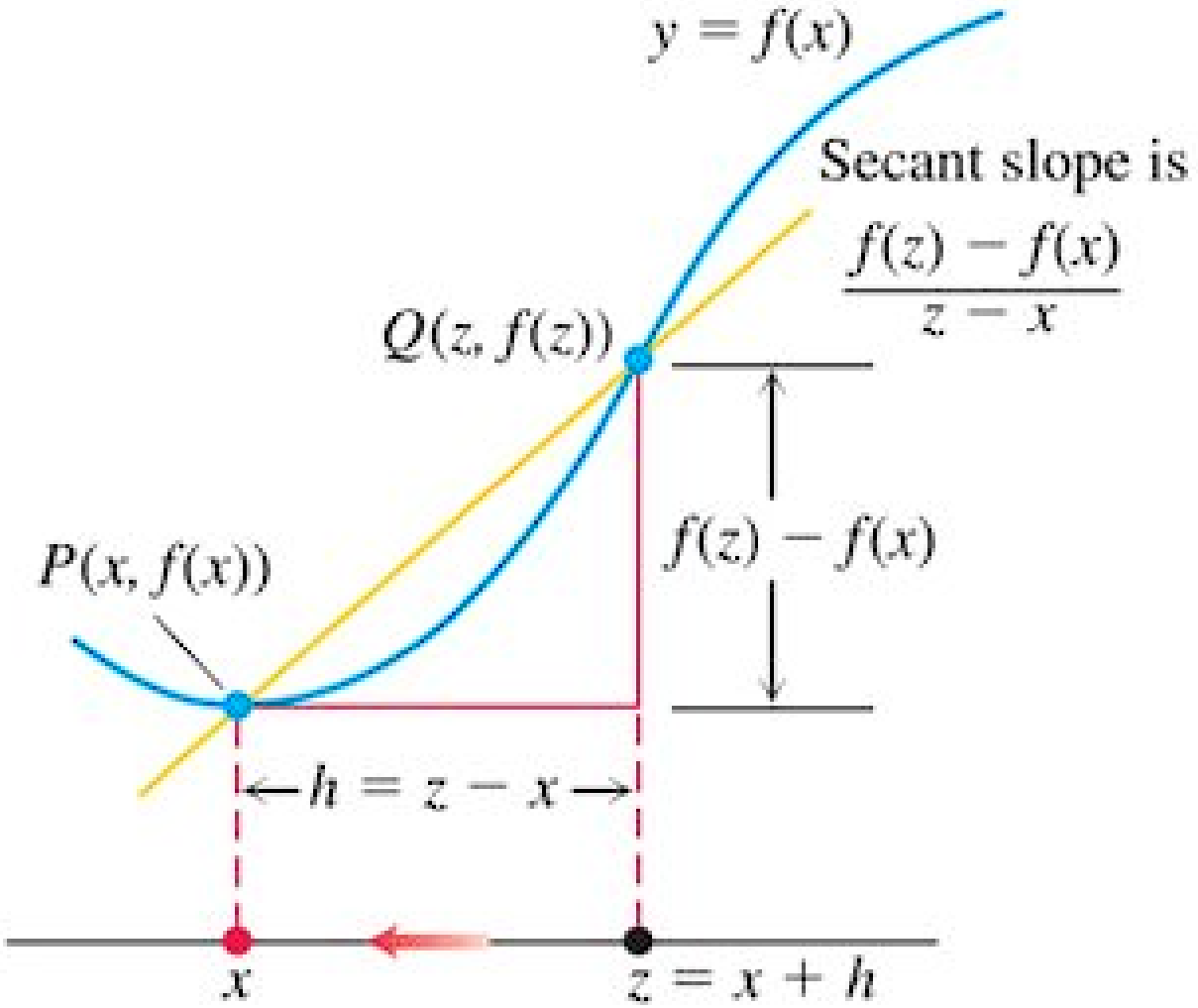
$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Alternatively,

$$f'(x) = \lim_{z \to x} \frac{f(z) - f(x)}{z - x}$$

provided the limit exists.

Definitions as a Graph



Examples

Example

Compute the derivatives of the following functions using the alternate definition:

- $f(x) = x^2$
- $f(x) = \sqrt{x}$
- $f(x) = \frac{1}{x}$

Notation

Definition

All of the following represent the derivative:

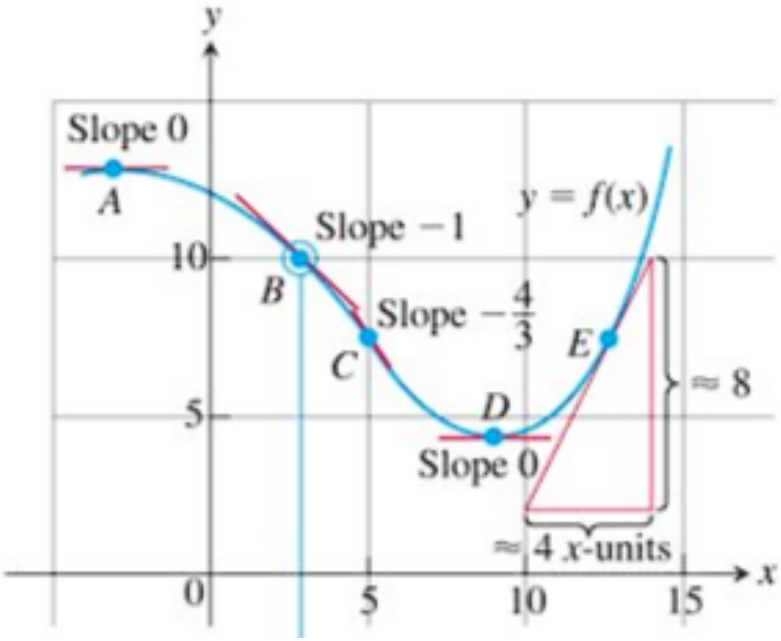
- $f'(x)$ (Newton)
- y' (Newton)
- $\frac{dy}{dx}$ (Leibniz)
- $\frac{d}{dx}f(x)$ (Leibniz)

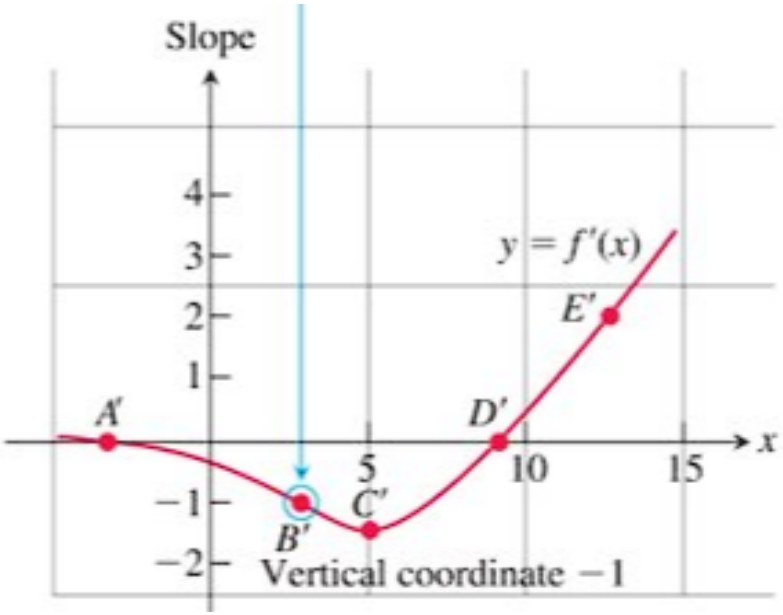
Furthermore, we use $f'(x_0)$ or $\left.\frac{dy}{dx}\right|_{x=x_0}$ to denote the derivative at a point.

Graphing $f'(x)$

Remark

Given the graph of $f(x)$, we can create the graph of $f'(x)$ by computing the slopes of tangent lines at various x values. These slopes become the y coordinates for the graph of $f'(x)$.

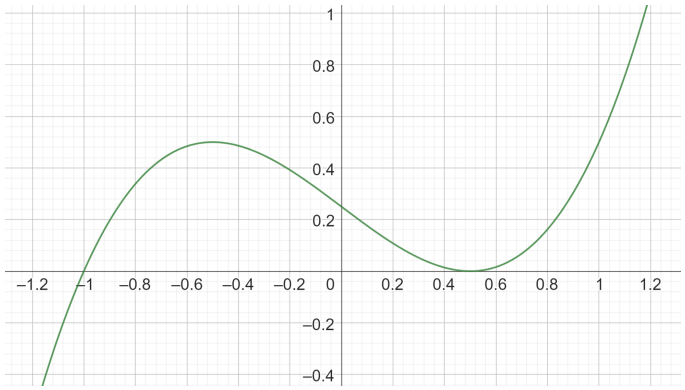


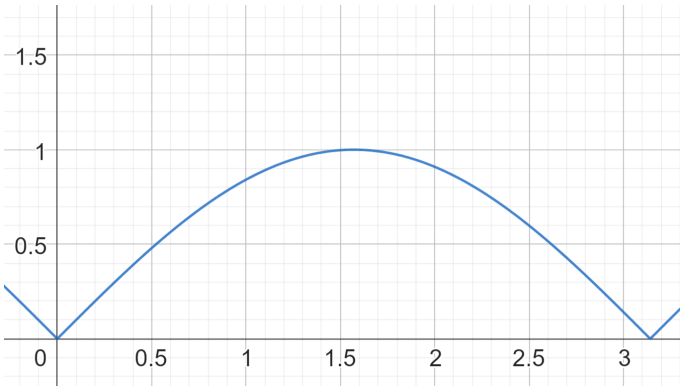


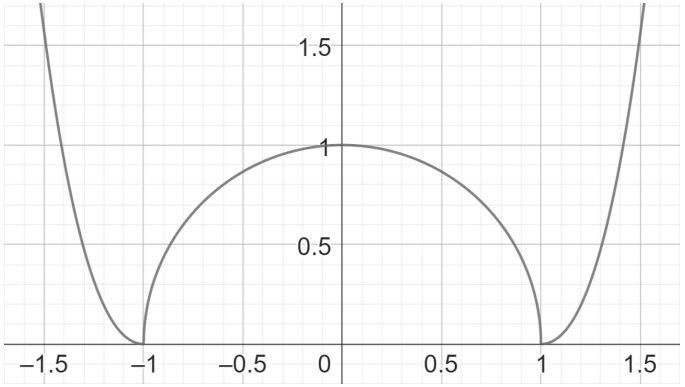
Examples

Example

For each given graph of $f(x)$, graph $f'(x)$.







Differentiable vs. Continuous

Definition

When a derivative exists at a point, we say the function $f(x)$ is **differentiable** at that point. This is normally characterized by "smoothness" in the function.

Theorem

If a function is differentiable at $x = x_0$, then it must be continuous at $x = x_0$. The converse of this statement (i.e. stated in the opposite order) is NOT necessarily true.

When Derivatives Do Not Exist

Remark

Derivatives do not exist when $f(x)$ has the following properties:

- *Corners, where derivatives differ from each side ($f(x) = |x|$)*
- *Cusps, where the function is continuous but one side has a derivative of ∞ and the other has a derivative of $-\infty$ ($f(x) = \sqrt{|x|}$)*
- *Vertical tangent lines, where the function is continuous and each side has the same infinite derivative ($f(x) = \sqrt[3]{x}$)*
- *Discontinuities ($f(x) = \frac{1}{x}$)*
- *Infinite oscillations ($f(x) = x\sin(1/x)$ when $x \neq 0$ and 0 when $x = 0$)*